

## RIMZENBERG PLC — COMPANY BRIEF

### Internal Onboarding Document — Junior Cloud Solutions Architect

#### 1. THE COMPANY

**Rimzenberg plc** is a UK-headquartered digital commerce and cloud solutions retailer, trading from offices in **Shoreditch, London**, with a secondary operations hub in **Leeds**. Rimzenberg sells home and lifestyle products through its own e-commerce platform and a network of 40+ third-party marketplace integrations.

- **Sector:** UK Digital Commerce & Cloud Solutions
- **Headcount:** 312 employees (38 in Technology, of which 3 are in the Cloud Engineering team you are joining)
- **Annual online revenue:** £64.2M
- **Hosting:** AWS (eu-west-2, London) — migrated from an on-premise data centre 18 months ago, but the migration was rushed and several architectural shortcuts were taken
- **Stock listing:** AIM-listed (smaller UK public company), which means Rimzenberg is subject to public-company governance expectations even though it is not in a heavily regulated sector like finance or healthcare

Rimzenberg's Technology division is small relative to its revenue. The Cloud Engineering team (you, plus two others) is expected to own architecture decisions that, at a larger company, would sit across multiple specialist teams.

#### 2. THE TRIGGER EVENT

**Black Friday, last year:** Rimzenberg's website went down for 4 hours and 12 minutes during the single highest-traffic period of the company's financial year. A single-AZ EC2 deployment behind a single load balancer could not absorb the traffic spike, the RDS database (sitting in a public subnet with no failover) became a bottleneck, and the team had no documented disaster recovery plan to fall back on.

**Estimated cost of the outage: £180,000 in lost revenue**, plus reputational damage reflected in a measurable dip in repeat-customer rate the following quarter.

The Board approved a cloud transformation budget the following week. You have been hired as **Rimzenberg's first dedicated Junior Cloud Solutions Architect** to help redesign the platform's architecture before it happens again — and before a scheduled external audit.

#### 3. THE HARD DEADLINE

Two deadlines anchor your work, both fixed and both non-negotiable:

| Deadline                                                           | Date                                                                     | What It Means                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NCSC / ISO 27001-aligned Security &amp; Architecture Review</b> | <b>Week 6 of your sprint</b>                                             | An external advisory firm reviews Rimzenberg's cloud security posture, IAM design, and network architecture against NCSC Cloud Security Principles and ISO 27001 controls. Failing this review delays the company's planned PCI-DSS-adjacent payment partner renewal. |
| <b>Black Friday Readiness Sign-Off</b>                             | <b>Week 10 of your sprint (12 weeks before this year's Black Friday)</b> | The Board requires written sign-off that the platform can survive a repeat of last year's traffic spike, with a documented, tested disaster recovery plan.                                                                                                            |

Your 8-phase work plan is scheduled across this 10-week sprint. Each phase maps to a named sprint week, shown in the table below.

| Phase                                           | Sprint Week | Milestone                                                       |
|-------------------------------------------------|-------------|-----------------------------------------------------------------|
| 1 — Discovery & Requirements                    | Week 1      | Requirements signed off by Engineering Manager                  |
| 2 — Highly Available Web Application            | Week 2      | HA architecture diagram reviewed                                |
| 3 — Three-Tier Cloud Architecture               | Week 3      | Three-tier design + IAM policy reviewed                         |
| 4 — Serverless Order Processing                 | Week 4      | Lambda/API Gateway flow deployed and demoed                     |
| 5 — Security, Networking & Identity             | Week 5–6    | <b>Must be complete before the Week 6 NCSC/ISO 27001 review</b> |
| 6 — Scalability, Monitoring & Cost Optimisation | Week 7      | Monitoring + cost report delivered to Finance                   |

|                                             |          |                                                             |
|---------------------------------------------|----------|-------------------------------------------------------------|
| 7 — Disaster Recovery & Business Continuity | Week 8-9 | DR plan tested ahead of Black Friday readiness sign-off     |
| 8 — Final Cloud Architecture Portfolio      | Week 10  | Board presentation — <b>Black Friday Readiness Sign-Off</b> |

## 4. CURRENT METRICS, TARGETS, AND THE COST OF THE GAP

| Metric                                        | Current State                                                                                                  | Target                                                                                | Cost / Risk of Gap                                                                         |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Web application availability during peak load | Single-AZ, no failover (effectively ~95% during incidents)                                                     | 99.9% (Multi-AZ, Auto Scaling, ALB)                                                   | £180,000 per major outage (Black Friday precedent)                                         |
| Database network exposure                     | RDS in public subnet, security group allows broad inbound access                                               | RDS in private subnet, tiered security groups, no direct public access                | Single point of compromise — an audit failure risk and a UK GDPR personal-data breach risk |
| Order processing compute cost                 | Always-on EC2 instances for an order workflow that spikes only at checkout (est. £4,100/month in idle compute) | Serverless (Lambda + API Gateway), pay-per-invocation                                 | Est. 60-70% reduction in compute cost for this workload if migrated                        |
| IAM access governance                         | No documented least-privilege model; several engineers hold AdministratorAccess                                | Documented IAM policy per role, KMS-encrypted secrets, audit trail via CloudTrail     | NCSC Cloud Security Principles failure risk at Week 6 review                               |
| Monthly AWS spend trend                       | Risen from £6,800/month to £21,400/month over two quarters with no monitoring or alerting in place             | Cost visibility via Cost Explorer + CloudWatch alerting, documented optimisation plan | Finance cannot currently explain or forecast cloud spend                                   |
| Disaster recovery readiness                   | No documented RTO/RPO, no tested failover, single-region                                                       | RTO ≤ 1 hour, RPO ≤ 15 minutes, Multi-AZ failover, cross-region S3 backup             | Repeat of last year's outage, or worse — a full regional AWS outage with no recovery plan  |

## 5. THE FOUR PEOPLE YOU WORK WITH

### Priya Nakamura — Engineering Manager (your line manager)

**Relationship: manager** Priya has been at Rimzenberg for four years and was promoted into the Engineering Manager role shortly after the Black Friday outage — she inherited the architecture problem rather than created it, and she is under direct pressure from the Board to show measurable progress. She is supportive but blunt about deadlines, and she will push back hard on any design that "looks good on paper" but cannot be demonstrated working. She signs off every phase before you can move to the next.

### Dev Okonkwo — Senior Cloud Engineer (outgoing colleague)

**Relationship: colleague** Dev built most of Rimzenberg's current AWS environment during the rushed migration 18 months ago and is leaving the company at the end of your second sprint week to join another employer. Dev's handover documentation is incomplete — covering only the basics of the current EC2/RDS setup — and Dev has quietly admitted to Priya that "there's stuff in there I never got round to documenting properly." Dev's messages contain genuine gaps and at least one buried hint about a configuration issue that becomes relevant later.

### Farah Hussain — Head of Operations

**Relationship: stakeholder** Farah owns the operational and commercial side of the business — she cares about uptime, customer experience, and cost, in that order, but she is currently most vocal about cost because Finance has been asking her hard questions about the cloud spend trend. She will sometimes push for the cheapest option even where it isn't the most resilient one, creating a natural tension with security/reliability priorities.

### Marcus Webb — Head of Security & Compliance (DPO)

**Relationship: stakeholder** Marcus is Rimzenberg's Data Protection Officer and owns the relationship with the external NCSC/ISO 27001-aligned auditors. He is the hard constraint in this simulation: any architecture decision that doesn't satisfy UK GDPR data protection requirements or NCSC Cloud Security Principles will be flagged by Marcus and must be addressed before the Week 6 review. Marcus communicates in precise, slightly formal language and always cites the specific principle or regulation he's referring to.

## 6. REGULATORY AND FRAMEWORK CONTEXT

Rimzenberg's architecture decisions must be defensible against the following UK-specific frameworks, auto-detected as relevant to a UK digital commerce business handling customer personal and payment-adjacent data:

- **UK GDPR** — particularly around where and how customer personal data is stored, encrypted, and backed up (relevant to RDS design in Phase 3, IAM/KMS design in Phase 5, and backup/DR design in Phase 7)

- **NCSC Cloud Security Principles** — the 14 principles published by the UK National Cyber Security Centre, used as the basis for the Week 6 external review (relevant to Phases 3, 5, and 7)
- **AWS Well-Architected Framework** — the five (now six) pillars (Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability) used as the internal design language for every Architecture Decision Log entry across Phases 2–7
- **ISO 27001** — the international information security management standard; Rimzenberg is pursuing alignment (not full certification) ahead of a future payment-partner renewal, so IAM, access logging, and audit-trail design in Phase 5 must map to relevant ISO 27001 Annex A controls

## 7. GLOSSARY — SHORTHAND AND TERMS USED IN LATER FILES

| Term / Code             | Meaning                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| <b>AZ</b>               | Availability Zone — an isolated AWS data centre location within a region                               |
| <b>ALB</b>              | Application Load Balancer                                                                              |
| <b>ASG</b>              | Auto Scaling Group                                                                                     |
| <b>RTO</b>              | Recovery Time Objective — maximum acceptable time to restore service after an outage                   |
| <b>RPO</b>              | Recovery Point Objective — maximum acceptable amount of data loss, measured in time                    |
| <b>SG</b>               | Security Group — a virtual firewall controlling inbound/outbound traffic at the instance level         |
| <b>NACL</b>             | Network Access Control List — a stateless firewall at the subnet level                                 |
| <b>IAM</b>              | Identity and Access Management                                                                         |
| <b>KMS</b>              | Key Management Service — used for encryption key management                                            |
| <b>CFN</b>              | Shorthand used internally at Rimzenberg for AWS CloudFormation                                         |
| <b>ADL</b>              | Architecture Decision Log — the required trade-off justification deliverable in every design phase     |
| <b>WAF (framework)</b>  | AWS Well-Architected Framework (not to be confused with AWS WAF, the web application firewall service) |
| <b>DPO</b>              | Data Protection Officer (Marcus Webb's role)                                                           |
| <b>P1 / P2 incident</b> | Internal Rimzenberg severity codes — P1 = full outage, P2 = degraded service                           |

*This brief is the starting reference point for every phase of the simulation. Process notes (File 2) will reference these personas, this glossary, and these metrics directly.*